

AA1 – 510 IN2 – Module 7 Project

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May 2025

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Obesity Level Prediction using Machine Learning

Introduction

Obesity is a growing global health concern linked to various chronic diseases, making its prediction and prevention critical for public health. This study leverages the Obesity Prediction Dataset, which combines real and synthetic data on lifestyle habits, eating behaviors, and physical activities, to develop predictive models for estimating obesity levels in individuals. By employing machine learning techniques, we aim to identify key factors influencing obesity and create accurate classification models. Our analysis seeks to provide insights into the relationships between dietary patterns, physical activity, and obesity, offering a foundation for targeted health interventions and personalized recommendations.

Problem Statement

Obesity is a complex condition influenced by a combination of genetic, environmental, and behavioral factors, such as eating habits and physical activity levels. The lack of precise, data-driven tools to predict obesity based on lifestyle factors hinders effective prevention and management strategies. The challenge lies in accurately estimating obesity levels using available data on individuals' habits and activities while identifying the most significant predictors. The Obesity Prediction Dataset provides an opportunity to address this by analyzing diverse features to predict obesity levels, but the integration of synthetic and real data introduces complexities in ensuring model robustness and generalizability.

Objective

The goal of this project is to predict obesity levels (e.g., Insufficient_Weight, Normal_Weight, Obesity_Type_I, etc.) using a Kaggle dataset containing features related to lifestyle, physical attributes,

and demographics. The project aims to demonstrate how machine learning using feature engineering and hyperparameter tuning can improve predictive performance, while also providing interpretable insights through evaluation metrics and visualizations.

Strategy

This project addresses the problem through the following objectives:

1. **Develop Predictive Models:** Build and evaluate machine learning models to accurately predict obesity levels based on features such as eating habits, physical activity, and demographic data from the Obesity Prediction Dataset.
2. **Identify Key Predictors:** Determine the most influential factors contributing to obesity through feature importance analysis, providing insights into lifestyle and behavioral drivers.
3. **Enhance Public Health Strategies:** Offer data-driven recommendations for targeted interventions by highlighting correlations between specific habits and obesity risk.
4. **Validate Model Robustness:** Assess the performance of predictive models on both real and synthetic data to ensure generalizability and reliability in real-world applications.
5. **Contribute to Personalized Healthcare:** Provide a foundation for developing personalized health recommendations by linking individual lifestyle patterns to obesity risk profiles.

By achieving these objectives, the study contributes to a broader understanding of obesity and supports the development of effective, evidence-based health interventions.

Methodology

1. Data Loading and Exploration

The Obesity Prediction Dataset, sourced from Kaggle, contains features such as Height, Weight, Gender, and lifestyle factors (e.g., frequency of physical activity, eating habits), with the target variable Obesity (renamed to Target) representing seven obesity levels: Insufficient_Weight, Normal_Weight, Overweight_Level_I, Overweight_Level_II, Obesity_Type_I, Obesity_Type_II, and Obesity_Type_III. The dataset combines real and synthetic data, necessitating careful validation to ensure model robustness.

- **Loading:** The dataset is loaded with error handling to verify file availability. The target column is renamed to Target for clarity.
- **Exploration:** Initial checks confirm no missing values, identify data types (numerical and categorical), and verify a balanced class distribution (approximately 60–70 samples per class in the test set). The synthetic data component raises concerns about potential noise or unrealistic patterns, which are addressed through robust preprocessing and validation.

2. Baseline Model Development

The baseline model establishes a reference for performance using minimal preprocessing and the XGBoost classifier, known for its efficiency in multiclass classification.

- **Preprocessing:**
 - Features (X_basic) include all columns except Target. This assumes all features are relevant, though domain knowledge highlights Height, Weight, and lifestyle factors as key predictors.
 - Categorical features (e.g., Gender, dietary habits) are encoded using LabelEncoder to convert them to numerical values.
 - The target variable is encoded, preserving class names for interpretability.
 - The dataset is split into 80% training and 20% testing sets, with stratification to maintain class balance and random_state=42 for reproducibility.

- **Training:** The XGBoost classifier uses default parameters with mlogloss as the evaluation metric, suitable for multiclass tasks.
- **Evaluation:**
 - Achieves ~95.51% accuracy, with precision, recall, and f1-scores ranging from 0.83 (Normal_Weight precision) to 1.00 (Obesity_Type_III precision).
 - The confusion matrix highlights minor misclassifications, particularly Normal_Weight confused with Overweight_Level_I, due to feature overlap.

3. Tuned Model with Feature Engineering

The tuned model enhances performance through feature engineering and hyperparameter optimization.

- **Feature Engineering:** Body Mass Index ($BMI = \text{Weight} / \text{Height}^2$) is added as a derived feature, leveraging its strong correlation with obesity levels to improve prediction accuracy.
- **Preprocessing:** Mirrors the baseline, with categorical encoding and consistent target encoding. The dataset is split identically for fair comparison.
- **Hyperparameter Tuning:**
 - A grid search evaluates 216 combinations across `n_estimators` (100, 200, 300), `max_depth` (3, 5, 8), `learning_rate` (0.05, 0.1, 0.2), `subsample` (0.8, 1.0), `colsample_bytree` (0.8, 1.0), and `gamma` (0, 0.1) using 5-fold cross-validation.
 - Best parameters: `colsample_bytree=0.8`, `gamma=0`, `learning_rate=0.2`, `max_depth=8`, `n_estimators=200`, `subsample=0.8`.
- **Evaluation:** Achieves ~98.58% accuracy, with near-perfect metrics (0.97–1.00) across classes, significantly improving Normal_Weight performance.

4. Model Evaluation and Validation

- **Metrics:** The tuned model's ~3% accuracy gain reflects the impact of BMI and tuning. The classification report shows balanced performance, with Normal_Weight precision improving from 0.83 to 0.98.
- **Confusion Matrices:** Side-by-side heatmaps reveal fewer misclassifications in the tuned model, particularly for Normal_Weight and Overweight_Level_I.
- **Feature Importance:** BMI, Weight, Height, and lifestyle factors (e.g., physical activity) are top predictors, aligning with Objective 2.
- **Robustness:** Stratified splitting and cross-validation ensure reliability. To address synthetic data concerns (Objective 4), performance is consistent across folds, but testing on a real-only subset could further validate generalizability.

5. Visualizations

- **Confusion Matrices:** Side-by-side heatmaps use the Blues colormap with annotated values, clearly showing improved predictions in the tuned model.
- **Feature Importance Plot:** A bar plot highlights BMI as a key predictor, alongside Weight and lifestyle factors, supporting Objective 3 by identifying actionable drivers.
- **Interpretability:** Visualizations are designed for accessibility, enabling healthcare professionals to understand model performance and key factors.

6. Model Saving

The tuned model is saved as `best_xgb_obesity_model.json`, facilitating deployment in health applications (Objective 5).

Results and Contributions

Objective 1: Develop Predictive Models

The baseline (95.51%) and tuned (98.58%) models demonstrate high accuracy, with the tuned model excelling due to BMI and optimized parameters.

Objective 2: Identify Key Predictors

Feature importance analysis confirms BMI, Weight, Height, and lifestyle factors as primary drivers, providing insights into obesity's behavioral and physical determinants.

Objective 3: Enhance Public Health Strategies

The analysis reveals correlations between low physical activity, high-calorie diets, and obesity risk, informing targeted interventions like exercise programs or dietary guidelines.

Objective 4: Validate Model Robustness

Consistent performance across cross-validation folds and balanced metrics suggest robustness, though synthetic data requires further validation on real-world datasets.

Objective 5: Contribute to Personalized Healthcare

The model's high accuracy and saved format enable integration into health apps or clinical tools, linking lifestyle patterns to personalized obesity risk profiles.

Practical Applications

- **Clinical Decision Support:** Assists doctors in identifying at-risk patients.
- **Health Apps:** Powers personalized risk assessments.
- **Public Health:** Informs population-level obesity prevention strategies.
- **Research:** Guides studies on lifestyle-obesity relationships.

Challenges and Considerations

- **Synthetic Data:** May introduce noise, requiring validation on real data.

- **Overfitting:** Deeper trees (max_depth=8) risk overfitting; additional regularization (e.g., lambda, alpha) could help.
- **Interpretability:** Feature importance is insightful, but tools like SHAP could enhance trust in medical settings.
- **Scalability:** Grid search is computationally intensive; randomized search or Bayesian optimization could improve efficiency.

Potential Enhancements

1. **Feature Engineering:** Add interaction terms (e.g., Weight $\times$ Age) or scale numerical features.
2. **Class Imbalance:** Apply SMOTE if imbalances are detected in larger datasets.
3. **Alternative Algorithms:** Compare with Random Forest or LightGBM.
4. **Explainability:** Use SHAP for instance-level insights.
5. **Data Quality:** Add checks for outliers or unrealistic values.
6. **Deployment:** Develop an API (e.g., Flask) for real-time predictions, ensuring compliance with regulations like HIPAA.

Conclusion

This project successfully leverages the Obesity Prediction Dataset to build robust XGBoost models, achieving 95.51% (baseline) and 98.58% (tuned) accuracy. The addition of BMI and hyperparameter tuning significantly enhances performance, particularly for challenging classes. Visualizations and feature importance analysis provide actionable insights into obesity drivers, supporting public health strategies and personalized healthcare. Future enhancements could further improve robustness and deployability, making this a valuable tool for obesity prevention and management.